

THE LEMNISCATIC BRIDGE CONSTANT G^* : SEVEN DERIVATIONS AND A QUADRATIC

ABSTRACT. We study the constant $G^* = \gamma_1^2/(\sqrt{2}\gamma_2^2) \approx 2.9587$, where $\gamma_1 = \Gamma(\frac{1}{4})$ and $\gamma_2 = \Gamma(\frac{1}{2})$, which we call the *lemniscatic bridge constant*. We present seven equivalent representations of G^* —from the lemniscate arc length, the Gamma function, the complete elliptic integral, the arithmetic-geometric mean, the Jacobi theta function at the self-dual nome, Watson’s lattice integral, and the L -function of the CM elliptic curve $E: y^2 = x^3 - x$ —demonstrating that G^* sits at a nexus of classical analysis, algebraic geometry, and number theory.

We then study the quadratic

$$x^2 - 16G^{*2}x + 16G^{*3} = 0,$$

whose coefficient 16 is the square of the order of the geometric automorphism group of E , and whose roots are $x_+ = 137.036\dots$ and $x_- = 3.024\dots$. In normalised form the harmonic mean of the roots is exactly $2 = [\mathbb{Q}(i) : \mathbb{Q}]$. We conjecture that x_+ equals α^{-1} , the inverse fine structure constant, and discuss the open problems that remain.

This paper is the first of three; the second [9] catalogues fifty-two representations of G^* , and the third [10] establishes G^* as the limit of a Wallis-type product over Fermat residue classes.

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1. INTRODUCTION

The fine structure constant

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036} \quad (1)$$

governs the strength of the electromagnetic interaction. Its value has been measured to extraordinary precision—the CODATA 2022 recommended value is $\alpha^{-1} = 137.035\,999\,177(21)$ —yet no accepted theory predicts it from first principles.

Over the decades, many authors have proposed closed-form expressions for α^{-1} in terms of mathematical constants. Wyler [15] used symmetric-space volumes to obtain $\alpha^{-1} \approx 137.036\,082$; others [6] employed group-theoretic constructions. These attempts share a common weakness: the expressions are chosen *ad hoc*, with no structural reason linking the mathematical objects to the electromagnetic coupling.

In this paper we take a different approach. We begin with a single elliptic curve—the unique rational curve with complex multiplication by the Gaussian integers and j -invariant 1728—and study the quadratic polynomial whose coefficients are intrinsic invariants of that curve. The larger root of this quadratic agrees with α^{-1} to 1.26 ppm. We make no claim that this agreement constitutes a derivation; we present it as a conjecture and identify the precise mathematical gaps that remain. Unlike previous attempts, the present construction is determined by the intrinsic arithmetic of a single CM curve—the only rational elliptic curve with j -invariant 1728 and complex multiplication by the Gaussian integers.

The central object is the *lemniscatic bridge constant*

$$G^* = \frac{\gamma_1^2}{\sqrt{2}\gamma_2^2} = 2.9586751191886388\dots, \quad (2)$$

where $\gamma_1 = \Gamma(\frac{1}{4})$ and $\gamma_2 = \Gamma(\frac{1}{2})$ are values of the Euler Gamma function at two rational arguments. The classical form $G^* = \sqrt{2}\Gamma(\frac{1}{4})^2/(2\pi)$ is algebraically equivalent, since $\pi = \gamma_2^2$. A comprehensive catalogue of fifty-two representations of G^* , spanning hypergeometric evaluations, modular forms, and lattice sums, is given in the companion paper [9].

This constant has been computed since Gauss [5] under various guises; we collect seven equivalent representations and prove they are all equal (Theorem 3.1).

2. THE CURVE AND ITS INVARIANTS

We collect the properties of the curve $E: y^2 = x^3 - x$ that enter the construction. All statements are classical.

Proposition 2.1 (CM structure). *The elliptic curve $E: y^2 = x^3 - x$ over $\mathbb{Q}$ satisfies:*

- (i) $j(E) = 1728$.
- (ii) $\text{End}_{\overline{\mathbb{Q}}}(E) \cong \mathbb{Z}[i]$ (complex multiplication by the Gaussian integers).
- (iii) The geometric automorphism group is $\text{Aut}_{\overline{\mathbb{Q}}}(E) = \{\pm 1, \pm i\} \cong \mathbb{Z}/4\mathbb{Z}$, with $|\text{Aut}_{\overline{\mathbb{Q}}}(E)| = 4$. Over $\mathbb{Q}$ itself, only $\text{Aut}_{\mathbb{Q}}(E) = \{\pm 1\}$ survives; the full group is defined over $\mathbb{Q}(i)$.
- (iv) $E(\mathbb{Q})_{\text{tors}} = \{O, (0, 0), (1, 0), (-1, 0)\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, with $|E(\mathbb{Q})_{\text{tors}}| = 4$.

(v) The conductor is $N = 32$ (LMFDB label **32.a2**) and the root number is $\varepsilon(E) = +1$.

Proof. See Silverman [13] IV.4 for (i)–(iv) and Cremona’s tables [3] for the conductor; the root number computation follows Rohrlich [11]. $\square$

Remark 2.2. The distinction between $\text{Aut}_{\overline{\mathbb{Q}}}(E)$ and $\text{Aut}_{\mathbb{Q}}(E)$ will matter: the number $16 = |\text{Aut}_{\overline{\mathbb{Q}}}(E)|^2$ that enters the master quadratic is the square of the *geometric* automorphism count. This is the same quantity that appears in the denominator of the BSD period ratio (cf. §2.1).

The CM field is $K = \mathbb{Q}(i)$, with $[K : \mathbb{Q}] = 2$.

2.1. The BSD computation. The Birch–Swinnerton-Dyer formula, proven for rank-zero CM curves by Rubin [12], gives

$$L(E, 1) = \frac{\Omega_+ \cdot |\text{III}| \cdot \prod_p c_p}{|E(\mathbb{Q})_{\text{tors}}|^2} \quad (3)$$

where $\Omega_+ = \varpi$ is the real period, $|\text{III}| = 1$ (Rubin [12]), $c_2 = 4$ is the Tamagawa number at $p = 2$ (the unique prime of bad reduction), $c_p = 1$ for all odd p , and $|E(\mathbb{Q})_{\text{tors}}| = 4$. Hence

$$L(E, 1) = \frac{\varpi \cdot 1 \cdot 4}{16} = \frac{\varpi}{4} \neq 0. \quad (4)$$

The nonvanishing was first proved by Coates–Wiles [4]. The representation $G^* = 8L(E, 1)/\gamma_2$ (Theorem 3.1, (11)) thus links G^* directly to the arithmetic of E via the BSD formula.

Remark 2.3. The denominator $|E(\mathbb{Q})_{\text{tors}}|^2 = 16$ in (3) is the same integer that will appear as the coefficient of the master quadratic. For this specific curve, this denominator also equals $|\text{Aut}_{\overline{\mathbb{Q}}}(E)|^2$ —a coincidence of small integers, not a general identity.

3. SEVEN REPRESENTATIONS OF G^*

Theorem 3.1 (Equivalence of representations). *The following seven expressions all equal G^* :*

$$G^* = \frac{2\varpi}{\gamma_2} \quad (\text{Lemniscate}) \quad (5)$$

$$G^* = \frac{\gamma_1^2}{\sqrt{2}\gamma_2^2} \quad (\text{Gamma function, } \pi\text{-free}) \quad (6)$$

$$G^* = \frac{2\sqrt{2}}{\gamma_2} K\left(\frac{1}{\sqrt{2}}\right) \quad (\text{Elliptic integral}) \quad (7)$$

$$G^* = \frac{2\gamma_2}{\text{agm}(1, \sqrt{2})} \quad (\text{Arithmetic-geometric mean}) \quad (8)$$

$$G^* = \gamma_2 \sqrt{2} \theta_3(e^{-\gamma_2^2})^2 \quad (\text{Theta function at self-dual nome}) \quad (9)$$

$$G^* = \sqrt{2\gamma_2^2 W_3} \quad (\text{Watson integral}) \quad (10)$$

$$G^* = \frac{8 L(E, 1)}{\gamma_2} \quad (L\text{-function of } E) \quad (11)$$

Here $\varpi = \gamma_1^2/(2\sqrt{2}\gamma_2)$ is the lemniscate constant, $K(k)$ the complete elliptic integral of the first kind, $\theta_3(q) = \sum_{n \in \mathbb{Z}} q^{n^2}$ the Jacobi theta function, and $W_3 = \gamma_1^4/(4\gamma_2^6)$ the Watson integral for the BCC lattice (Watson [14]; see also Zucker–Joyce [16]). An extended catalogue of forty-five further representations is given in [9].

Proof. Each reduces to (6) by a single classical identity:

- (5): $\varpi = G^* \gamma_2/2$ by definition.
- (7): Legendre [7] evaluates $K(1/\sqrt{2}) = \gamma_1^2/(4\gamma_2)$.
- (8): Gauss [5] proved $\text{agm}(1, \sqrt{2}) = \gamma_2^2/\varpi$ (see also Borwein–Borwein [1]).
- (9): The Chowla–Selberg formula [2] gives $\theta_3(e^{-\gamma_2^2}) = \gamma_1/(\gamma_2^{3/2} \cdot 2^{1/4})$.
- (10): Watson [14] proved $W_3 = \gamma_1^4/(4\gamma_2^6)$; then $2\gamma_2^2 W_3 = \gamma_1^4/(2\gamma_2^4) = G^{*2}$.
- (11): $L(E, 1) = \varpi/4$ by (4). □

4. THE TRIAD IDENTITY

Theorem 4.1 (Triad invariant). $\pi = 4\varpi^2/G^{*2}$.

Proof. $G^* = 2\varpi/\gamma_2$ and $\pi = \gamma_2^2$. □

5. THE MASTER QUADRATIC

We now state the central conjectural construction.

Conjecture 5.1 (Master quadratic). *The polynomial*

$$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3} \quad (12)$$

is the characteristic polynomial of a canonically defined self-pairing on the CM curve E : $y^2 = x^3 - x$, with $16 = |\operatorname{Aut}_{\overline{\mathbb{Q}}}(E)|^2$ and G^* the scaled period.

The polynomial P is unambiguously defined: its coefficients are built from G^* and the integer 16. What is conjectural is the claim that P arises canonically from the geometry of E . We discuss the evidence for each ingredient.

5.1. Why degree 2? The CM field $K = \mathbb{Q}(i)$ has $[K : \mathbb{Q}] = 2$. Every endomorphism $\varphi \in \operatorname{End}(E)$ satisfies a degree-2 characteristic polynomial $t^2 - \operatorname{Tr}(\varphi)t + \deg(\varphi) = 0$ over $\mathbb{Z}$. It is therefore natural—though not logically forced—that a “global” coupling polynomial should also have degree 2.

Question 5.2. Is there a rigorous argument showing that the self-consistent coupling of E on $\mathbb{Z}^3$ must satisfy a polynomial of degree exactly $[K : \mathbb{Q}]$?

5.2. Why coefficient 16? The integer $16 = |\operatorname{Aut}_{\overline{\mathbb{Q}}}(E)|^2 = |E(\mathbb{Q})_{\text{tors}}|^2$ appears in the BSD formula (3) as the denominator of the period ratio. It is the natural “normalisation divisor” for this curve. We conjecture that it enters the master quadratic for the same structural reason.

Question 5.3. Does the BSD denominator $|E(\mathbb{Q})_{\text{tors}}|^2$ canonically determine the coefficient of a self-consistency equation on E ?

5.3. Why Sum = Product? In the normalised variable $u = x/G^*$, the quadratic becomes

$$u^2 - S u + S = 0, \quad S = 16 G^*. \quad (13)$$

The distinctive feature is that the linear coefficient and the constant term are *equal*. By Vieta:

$$u_+ + u_- = u_+ \cdot u_- = S. \quad (14)$$

Theorem 5.4 (Harmonic mean). *For any quadratic $u^2 - Su + S = 0$ with $S \neq 0$, the harmonic mean of the roots is*

$$H = \frac{2 u_+ u_-}{u_+ + u_-} = \frac{2S}{S} = 2. \quad (15)$$

Proof. Immediate from (14). □

The coincidence $H = 2 = [K : \mathbb{Q}]$ is the strongest structural evidence for (12): if one accepts that the polynomial should be degree 2 (matching $[K : \mathbb{Q}]$) and should have Sum = Product (giving $H = [K : \mathbb{Q}]$), then (12) is the unique result.

Conjecture 5.5 (Self-duality forces Sum = Product). *The root number $\varepsilon(E) = +1$ makes the functional equation $\Lambda(E, s) = \Lambda(E, 2 - s)$ an identity at $s = 1$. We conjecture that this self-duality, when applied to a suitably defined “global coupling operator” on the Galois eigenspaces of $T_0(E)$, forces the operator’s trace and determinant to be equal.*

Remark 5.6. Conjecture 5.5 is the central open problem. The “global coupling operator” is not yet rigorously defined. Candidate constructions include the Petersson self-pairing of the weight-2 newform $f \in S_2(\Gamma_0(32))$ associated to E , decomposed by the $\operatorname{Gal}(K/\mathbb{Q})$ action. We do not pursue this further here.

5.4. Roots.

Theorem 5.7 (Explicit roots). *The roots of (12) are*

$$x_+ = 8G^{*2} + 4G^* \sqrt{G^*(4G^* - 1)} = 137.0361714582 \dots, \quad (16)$$

$$x_- = 8G^{*2} - 4G^* \sqrt{G^*(4G^* - 1)} = 3.0239639163 \dots. \quad (17)$$

*The discriminant is $\Delta = 64G^{*3}(4G^* - 1) > 0$. Both roots are transcendental: since γ_1 and $\gamma_2^2 = \pi$ are algebraically independent over $\mathbb{Q}$ (Nesterenko [8]), G^* is transcendental, and a quadratic in G^* with integer-ratio structure cannot produce algebraic roots.*

Proof. Quadratic formula. The transcendence claim follows from Nesterenko [8], Theorem 1. $\square$

Corollary 5.8. $\lfloor x_- \rfloor = 3$.

5.5. The k -family. Replacing 16 by a parameter k , the normalised quadratic $u^2 - kG^*u + kG^* = 0$ still has $H = 2$ for every $k > 0$. The discriminant vanishes at $k_{\text{crit}} = 4/G^* \approx 1.352$; below this value the roots are complex.

6. THE DUAL CONVERGENCE

Theorem 6.1 (Dual reading of r_2). *Let $r_2(n) = \#\{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 = n\}$ and $R(N) = \sum_{n=0}^N r_2(n)$. Then:*

- (i) $R(N)/N \rightarrow \pi$ as $N \rightarrow \infty$ (Gauss circle theorem).
- (ii) $\sum_{n=0}^{\infty} r_2(n) e^{-\gamma_2^2 n} = G^*/(\gamma_2 \sqrt{2})$ (Chowla–Selberg).

The same function r_2 produces π by unweighted averaging and G^ by exponential weighting at the self-dual nome.*

7. THE GAMMA-PRIMITIVE BASIS

Theorem 7.1 (Two-primitive basis). *Every object in the elliptic hierarchy at the lemniscatic modulus $k = 1/\sqrt{2}$ is a rational expression in γ_1 , γ_2 , and $\sqrt{2}$:*

$$\varpi = \frac{\gamma_1^2}{2\sqrt{2}\gamma_2}, \quad K(1/\sqrt{2}) = \frac{\gamma_1^2}{4\gamma_2}, \quad (18)$$

$$\text{agm}(1, \sqrt{2}) = \frac{2\gamma_2^3}{\gamma_1^2}, \quad E(1/\sqrt{2}) = \frac{8\gamma_2^4 + \gamma_1^4}{8\gamma_1^2\gamma_2}. \quad (19)$$

Proof. Legendre evaluations at $k = 1/\sqrt{2}$ (cf. Borwein–Borwein [1]). The E form follows from the Legendre relation: at $k = 1/\sqrt{2}$ we have $K = K'$ and $E = E'$, so $2KE - K^2 = \gamma_2^2/2$. $\square$

Corollary 7.2. *At $k = 1/\sqrt{2}$: (i) $E = (K + \text{agm}(1, \sqrt{2}))/2$; (ii) $E/K = 1/2 + 2/G^{*2}$; (iii) $\Pi(1/2, 1/\sqrt{2}) = 2E$.*

8. THE CONJECTURE

Conjecture 8.1 (Lattice-CM). *The inverse fine structure constant equals the larger root of the master quadratic:*

$$\alpha^{-1} = x_+ = 8G^{*2} + 4G^* \sqrt{G^*(4G^* - 1)}.$$

Numerical evidence. $x_+ = 137.0362\dots$ vs. CODATA $\alpha^{-1} = 137.0360(2)$. Agreement: 1.26 ppm, with zero adjustable parameters in the leading term.

Structural observations.

- $\lfloor x_- \rfloor = 3$, the number of spatial dimensions and of quark colour charges.
- $D = 3$ is the unique dimension $D \in \{1, \dots, 6\}$ for which the analogous quadratic (with the D -dimensional Watson integral) satisfies $\lfloor x_-^{(D)} \rfloor = D$.

Remark 8.2 (Running of α). The electromagnetic coupling runs with energy scale; $\alpha^{-1} \approx 137$ is the low-energy (infrared) value. If Conjecture 8.1 is correct, x_+ should be identified with this infrared limit. The connection to the renormalization group flow is an open problem.

Remark 8.3 (Comparison with prior work). Wyler [15] obtained $\alpha^{-1} \approx 137.03608$ from symmetric-space volumes; the expression has no structural link to an elliptic curve and the agreement is weaker. The present approach differs in that all ingredients— G^* , 16, the degree 2—are intrinsic invariants of a single CM curve. Whether this structural coherence elevates the result beyond coincidence remains to be established.

Remark 8.4 (Precision formula). We note that the expansion $\alpha^{-1} = x_+ - \frac{9}{47}|\varepsilon| + \frac{5}{64}|\varepsilon|^2 - \dots$ with $\varepsilon = e^\pi - \pi - 20 \approx -9 \times 10^{-4}$ matches CODATA to many digits. However, the rational coefficients in this expansion are *observed*, not derived, and the expansion variable is chosen because e^π is a famous near-integer. We include this observation for completeness; it should not be taken as evidence for Conjecture 8.1.

9. OPEN PROBLEMS

- (1) **The global coupling operator.** Does there exist a canonically defined operator on the motive of E whose characteristic polynomial is $P(x)$ as in (12)? (See Conjecture 5.1.)
- (2) **Why Sum = Product?** Does the self-duality $\Lambda(E, s) = \Lambda(E, 2 - s)$ at $s = 1$ force the trace and determinant of a coupling operator on E to be equal? (See Conjecture 5.5.)
- (3) **Why 16?** Is there a functorial construction that selects $16 = |\text{Aut}_{\overline{\mathbb{Q}}}(E)|^2$ as the coefficient of both the BSD period ratio and the master quadratic, or is the coincidence accidental?
- (4) **Physical mechanism.** Can lattice gauge theory on $\mathbb{Z}^3$, or any other physical framework, produce Conjecture 8.1 as a consequence of a dynamical principle?
- (5) **Running coupling.** If $x_+ = \alpha^{-1}$ at zero momentum transfer, what determines the beta function $\beta(\alpha)$ within this framework? A complete theory must predict the running, not just the infrared fixed point.

10. CONCLUSION

The lemniscatic bridge constant $G^* = \gamma_1^2/(\sqrt{2}\gamma_2^2)$ has been computed since Gauss. It admits seven equivalent representations spanning classical analysis, lattice combinatorics, and arithmetic geometry; a further forty-five appear in the companion catalogue [9], and a Wallis-type product formula is established in [10].

The quadratic $x^2 - 16G^{*2}x + 16G^{*3} = 0$, with coefficient $16 = |\text{Aut}_{\overline{\mathbb{Q}}}(E)|^2$ and normalised harmonic mean $H = 2 = [\mathbb{Q}(i) : \mathbb{Q}]$, has a larger root of $137.036\dots$. Whether this root is the inverse fine structure constant is a conjecture. We have identified three precise mathematical questions (the global coupling operator, the Sum = Product condition, and the role of 16) whose resolution would settle the matter.

We leave the question open and offer the bridge for crossing.

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